

Stat 17 Week 3

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Probability Topics — Part 1: Basics & Counting

- **What is Probability**

- Probability describes the long-run proportion of times an outcome occurs
- Probabilities are always between 0 and 1
 - * $P = 0$: impossible event
 - * $P = 1$: certain event

- **Small Samples vs Large Samples**

- Results from small samples can be misleading
- Short-term randomness does not reflect long-term behavior

- **Law of Large Numbers**

- As the number of trials increases, observed proportions approach theoretical probabilities
- Early imbalances tend to stabilize over time

- **Independence and the Gambler's Fallacy**

- Trials are independent if previous outcomes do not affect future outcomes
- Past results do not change future probabilities
- The belief that an outcome is “due” is incorrect

- **Probability Models and Sample Space**

- A probability model consists of:
 - * a list of all possible outcomes
 - * a probability assigned to each outcome
- The sample space S is the set of all possible outcomes

- **Multi-step Experiments**

- Experiments with multiple stages can be represented using tree diagrams
- Example: two coin tosses $\rightarrow \{HH, HT, TH, TT\}$

- **With Replacement vs Without Replacement**

- With replacement: the number of possible outcomes remains the same each trial
- Without replacement: the number of possible outcomes decreases after each trial

- **Order of Selection**

- If order matters, different sequences count as different outcomes
- If order does not matter, only the selected elements matter

- **Permutations (Order Matters)**

$${}_nP_r = \frac{n!}{(n-r)!} = n(n-1)\cdots(n-r+1)$$

- **Combinations (Order Does Not Matter)**

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\cdots(n-r+1)}{r!}$$